



Green University of Bangladesh

*Department of Computer Science and Engineering (CSE)
Semester: (Summer, Year: 2026), B.Sc. in CSE (Day)*

Digital Signal Encoding Simulator

*Course Title: Data Communication Lab
Course Code: CSE312
Section: 232 D3*

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<u>Lab Project Status</u>	
Marks:	Signature:
Comments:	Date:

Contents

1		2
1.1	Introduction	2
1.1.1	Overview	2
1.1.2	Motivation	2
1.1.3	Problem Definition	3
1.1.4	Objectives	3
1.1.5	Scope of the Project	3

Chapter 1

1.1 Introduction

1.1.1 Overview

The “Digital Signal Encoding Simulator” is a desktop-based educational tool designed to visualize and analyze digital-to-digital line encoding schemes. In data communication, converting digital data into digital signals (line coding) is a fundamental process at the Physical Layer of the OSI model. This application, built using Python and the Tkinter GUI framework, accepts either raw bit sequences or plain text (automatically converted to 8-bit ASCII binary) and dynamically renders the corresponding waveforms.

The simulator supports five critical line encoding techniques: Non-Return-to-Zero Level (NRZ-L), Non-Return-to-Zero Invert (NRZ-I), Manchester, Differential Manchester, and Alternate Mark Inversion (AMI). By providing a real-time, interactive visual representation of how bits translate to voltage levels, the system bridges the gap between abstract theoretical concepts and practical signal analysis.

1.1.2 Motivation

Understanding line coding is crucial for Computer Science and Electrical Engineering students, as it dictates how data is physically transmitted over a medium. However, learning these concepts purely through textbooks is challenging. Students often struggle to manually draw waveforms, understand mid-bit transitions, or visualize the effects of baseline wandering and DC components.

The motivation behind this project is to develop a simple, scalable, and educational desktop application that automates the waveform generation process. It allows users to instantly see the physical representation of bitstreams, making it easier to compare the synchronization capabilities, bandwidth requirements, and DC component characteristics of different encoding schemes. Furthermore, it serves as a practical exercise in algorithmic state-machine design and 2D canvas rendering.

1.1.3 Problem Definition

Problem Statement

Traditional methods of learning digital signal encoding rely heavily on manual diagramming, which is time-consuming and prone to human error, especially for long bit sequences. While some web-based simulators exist, they often require internet connectivity, suffer from browser rendering latency, or lack the ability to handle text-to-binary conversion seamlessly. Additionally, understanding the state-dependent nature of certain encodings (like NRZ-I and Differential Manchester) requires tracking previous bit states, which is difficult to visualize statically. This project aims to solve these problems by developing an offline, efficient, and interactive desktop application that accurately renders complex, state-dependent signal waveforms in real-time.

1.1.4 Objectives

The primary and secondary objectives of this project are:

- **Primary:** To develop a fully functional desktop simulator that accurately draws NRZ-L, NRZ-I, Manchester, Differential Manchester, and AMI waveforms.
- **Secondary:** To implement a text-to-binary converter that allows users to input plain text and view the corresponding encoded signals.
- **Secondary:** To provide a clear visual distinction between byte boundaries and individual bits to aid in educational comprehension.
- **Secondary:** To ensure the application is lightweight, requires no external dependencies (using only the Python Standard Library), and runs offline.

1.1.5 Scope of the Project

The scope of this project is strictly limited to digital-to-digital line encoding. It does not cover analog-to-digital (PCM) or digital-to-analog (ASK, FSK, PSK) modulation. The application is designed for educational and simulation purposes and does not interface with actual hardware network interfaces or physical transmission media.